Detailed Theory: IP Subnetting & Variable Length Subnet Masking (VLSM)

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1. What is IP Subnetting?

IP subnetting is the process of dividing a large network (IP address block) into smaller, more manageable subnetworks, or subnets. This improves network organization, increases routing efficiency, reduces broadcast traffic, and enhances security.

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2. Why Subnet?

- **Efficient IP address allocation:** Subnetting avoids wasting IP addresses by matching subnet sizes to actual needs.
- **Reduced broadcast domains:** Each subnet creates a separate broadcast domain, reducing network congestion.
- **Improved network management:** Smaller networks are easier to manage, troubleshoot, and secure.

- **Routing simplification:** Routers deal with subnet routes rather than many individual IP addresses.

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3. Basic Concepts

- **Network ID:** The portion of the IP address that identifies the specific network.
- **Host ID:** The portion of the IP address that identifies a host (device) on the network.
- **Subnet mask:** A 32-bit number that masks the IP address, separating the network and host portions.

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4. Subnet Mask

A subnet mask uses binary 1s to cover the network bits and 0s for host bits.

Example:

- For class C default subnet mask: 255.255.255.0 (binary: 11111111.11111111.11111111.00000000)

The subnet mask determines how many bits are used for network vs host identification.

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5. Calculating Subnets and Hosts

Given:

- Number of bits borrowed for subnetting = **n**
- Number of subnetworks = 2^n
- Number of hosts per subnet = 2^(remaining bits) - 2
  (The subtraction of 2 accounts for the network and broadcast addresses, which cannot be assigned to hosts.)

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6. Example: Subnetting a Class C Network

- Original network: 192.168.1.0 with default subnet mask 255.255.255.0
- Borrow 3 bits for subnetting: subnet mask becomes 255.255.255.224 (binary: 11111111.11111111.11111111.11100000)
- Number of subnets: 2^3 = 8
- Number of hosts per subnet: 2^(32-27) - 2 = 2^5 - 2 = 30 hosts per subnet

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7. Variable Length Subnet Masking (VLSM)

VLSM is a technique where different subnet masks are used within the same network to create subnets of varying sizes. This allows for more efficient IP address utilization compared to fixed-length subnetting.

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8. Why Use VLSM?

- Allows subnetting a subnet to accommodate networks with different host requirements.
- Helps reduce wasted IP addresses.
- Enables hierarchical network design.

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9. How VLSM Works

- You start by allocating subnets with the largest host requirement first.
- Use the smallest subnet mask (largest subnet) for these large subnets.

- Continue subnetting the remaining address space with smaller subnet masks for smaller networks.

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10. Example of VLSM

Suppose you have 192.168.1.0/24 and need:

- Subnet A: 100 hosts
- Subnet B: 50 hosts
- Subnet C: 10 hosts

Steps:

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For 100 hosts: Need 7 bits for hosts → subnet mask /25 (255.255.255.128) → 126 hosts per subnet

- Assign 192.168.1.0/25 to Subnet A
- Remaining addresses: 192.168.1.128/25
- For 50 hosts: Need 6 bits → subnet mask /26 (255.255.255.192) → 62 hosts per subnet
- Assign 192.168.1.128/26 to Subnet B
- Remaining addresses: 192.168.1.192/26
- For 10 hosts: Need 4 bits → subnet mask /28 (255.255.255.240) → 14 hosts per subnet
- Assign 192.168.1.192/28 to Subnet C

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11. CIDR Notation

Classless Inter-Domain Routing (CIDR) notation represents the IP address and subnet mask compactly as:

IP_address / prefix_length

Example: 192.168.1.0/24 means subnet mask 255.255.255.0

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12. Important Points

- Always subtract 2 from host bits for network and broadcast addresses.
- VLSM can only be used if routers support CIDR.
- VLSM enhances efficient utilization of IP address space.
- Proper planning is essential to avoid overlapping subnets.